MCV4U1-QUIZ Name________________

Formatvie Quiz: Derivatives of Exponential & Logarithmic Functions

K/U	A	C	TIPS
/16	/8	/6	/8

KNOWLEDGE / UNDERSTANDING

1. Find the first derivative: (fully factored with positive exponents)

a) $y=(3x-5)e^{3x}$ ✓

$\frac{dy}{dx}=3e^{3x}+e^{3x}(3)(3x-5)$

$=3e^{3x}(1+3x-5)$

$=3e^{3x}(3x-4)$ ✓

b) $y=\log_6(5x)$ ✓

$\frac{dy}{dx}=\frac{5}{5x\ln 6}$

$=\frac{1}{x\ln 6}$ ✓

c) $y=\ln(2x^2-1)^3$ ✓✓

$\frac{dy}{dx}=\frac{3(2x^2-1)^2(4x)}{(2x^2-1)^3}$ ✓

$=\frac{12x}{(2x^2-1)}$ ✓

d) $y=\ln(\ln x^2)$ ✓✓

$\frac{dy}{dx}=\frac{\frac{1}{x^2}\cdot 2x}{\ln x^2}$ ✓

$=\frac{2}{x\ln x^2}$ ✓

$=\frac{1}{x\ln x}$

e) $y=5^{2x-4}(2x-4)$ ✓✓

$\frac{dy}{dx}=5^{2x-4}(2)(\ln 5)(2x-4)$ ✓

$+(2)(5^{2x-4})$ ✓

$=2(5^{2x-4})[(2x-4)\ln 5+1]$

f) $y=e^{\sin x^2}$ ✓✓

$y'=e^{\sin x^2}(\cos x^2)(2x)$

g) $f(x)=\cos(x+1)\sin^2(x^2-3x)$ ✓✓✓

$f'(x)=-\sin(x+1)\sin^2(x^2-3x)$ ✓

$+2[\sin(x^2-3x)][\cos(x^2-3x)](2x-3)[\cos(x+1)]$ ✓

$=\sin(x^2-3x)[(-\sin(x+1))(\sin(x^2-3x))$ ✓

$+2[\cos(x^2-3x)](2x-3)]\cos(x+1)$.

h) $y=x^3\log_3(x^3-2x+3)$ ✓✓✓

$y'=3x^2\log_3(x^3-2x+3)$ ✓

$+\frac{3x^2-2}{(x^3-2x+3)\ln 3}\cdot x^3$ ✓

$=x^2\left[3\log_3(x^3-2x+3)+\frac{3x^3-2x}{(x^3-2x+3)\ln 3}\right]$ ✓

APPLICATION

2. Find the equation of the tangent to the curve $y=3^x$ at the point where $x=-2$ ✓✓✓✓

$\frac{dy}{dx}=3^x\ln 3$

at $x=-2$

$\frac{dy}{dx}=3^{-2}\ln 3$

$=\frac{\ln 3}{9}$

at $x=-2$

$y=3^{-2}$

$=\frac{1}{9}$

Equation of Line.

$y=mx+b$

$\frac{1}{9}=\frac{\ln 3}{9}(-2)+b$

$\frac{1}{9}+\frac{2\ln 3}{9}=b$

$b=\frac{1+2\ln 3}{9}$

$\therefore y=\frac{\ln 3}{9}x+\frac{1+2\ln 3}{9}$

MCV4U1-QUIZ Name_______________

3. Find the equation of the tangent to the curve $y = \ln(1+e^x)$ which is parallel to the line x - 2y = 0. ✓✓✓✓

$x - 2y = 0$
$2y = x$
$y = \frac{1}{2}x$
$\therefore$ Slope $= \frac{1}{2}$

$y = \ln(1+e^x)$
$\frac{dy}{dx} = \frac{e^x}{1+e^x}$
$\therefore m = \frac{1}{2}$.
$\frac{1}{2} = \frac{e^x}{1+e^x}$
$1+e^x = 2e^x$
$1 = e^x$
$x = 0$

at $x = 0$
$y = \ln(1+e^0)$
$y = \ln 2$
$\therefore$ Equation $y = mx + b$
$\ln 2 = \frac{1}{2}(0) + b$
$b = \ln 2$.
$\therefore y = \frac{1}{2}x + \ln 2$.

COMMUNICATION

4. a) State how to find the derivative of $y = a^{f(x)}$ in your own words. ✓✓

Multiply the derivative of f(x) with ln of 'a' and then with the original function y.

b) State how to find the derivative of $y = e^{f(x)}$ in your own words. ✓✓

Multiply the derivative of f(x) with ln of 'e', then with the function $e^{f(x)}$ to find the derivative of y

c) Explain how the rule in b) is a special case of the rule in a). ✓✓

The rule in b is the same as the rule a since ln e is equal to 1.

THINKING, INQUIRY AND PROBLEM SOLVING

5. The function $f(t) = 2000e^{3t}$ represents the number of bacteria after t hours. Find the rate of change of growth when the population of bacteria is at 800,000. ✓✓✓✓ rate of growth → 1st der. 2nd deriv.

Find t at population 800,000

$800000 = 2000e^{3t}$
$400 = e^{3t}$
$\ln 400 = \ln e^{3t}$
$\ln 400 = 3t$
$t = \frac{\ln 400}{3}$
$\doteq 2$. hours.

If this was only first der. then it would read "... change of growth..."

$f'(t) = 2000e^{3t}(3)$
$= 6000e^{3t}$

$f''(t) = 18\,000e^{3t}$
$f''(2) \doteq 18\,000e^{3(2)}$
$= 7261\,718$ bac/hr.

6. If $y = x^2e^x$ prove that $y''' = e^x(x^2 + 6x + 6)$ ✓✓✓✓

$y' = 2xe^x + e^x x^2$
$y'' = 2e^x + e^x(2x) + e^x x^2 + 2xe^x$
$= 2e^x + 4xe^x + e^x x^2$

$y''' = 2e^x + 4e^x + e^x(4x) + e^x x^2 + 2xe^x$
$= 6e^x + 6xe^x + e^x x^2$
$= e^x(6 + 6x + x^2)$